LumindAd

TelecomX Churn Intelligence Platform v1.0

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Python TensorFlow XGBoost SHAP MLP LSTM CNN-1D Autoencoder Folium

Customers	Churn Rate	Countries	Regions	Models	Best F1	Best AUC	Green AI
7,267	25.7%	54	6	5	0.6171	0.8252	YELLOW

Pipeline Overview

raw/TelecomX_Data.json [Flatten + Encode + Global Enrichment] processed/LumindAd_TelecomX_Global_v3.csv [SMOTE + RobustScaler] 5 ML Models [SHAP + Calibration] BI Reports

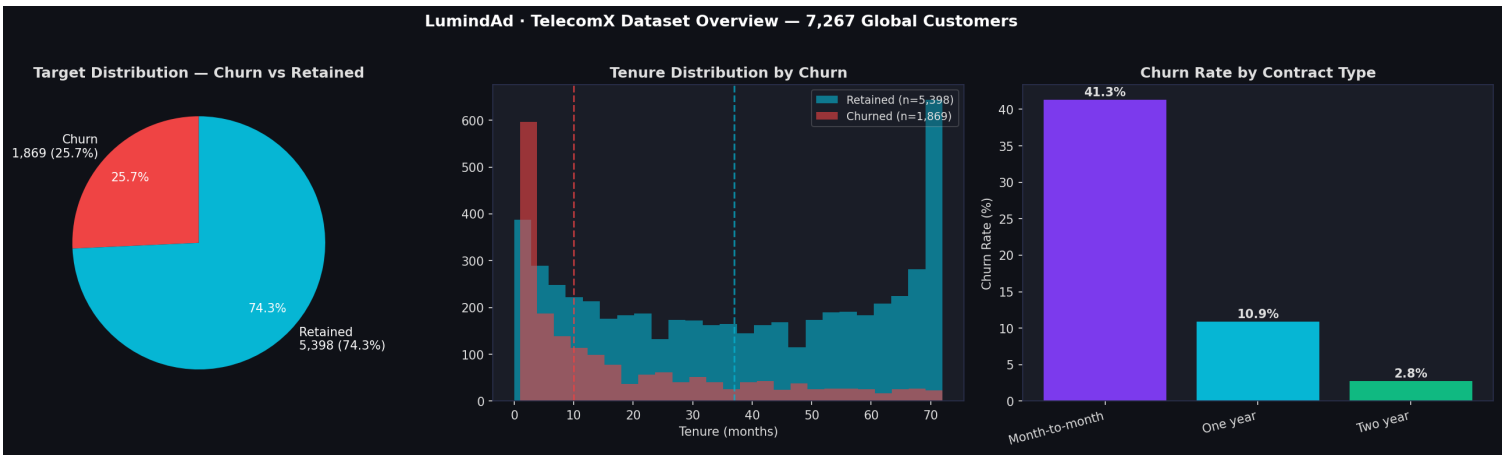
Global enrichment assigns country, region, lang, cpc_index for 54 countries in 7 regions. SMOTE balances the 74.3%/25.7% class imbalance. RobustScaler handles outliers in MonthlyCharges and TotalCharges. All DL thresholds calibrated via ROC F1-maximisation.

1. Dataset Overview -- LumindAd_TelecomX_Global_v3.csv

Source	TelecomX_Data.json (Alura LATAM Challenge)
Total Customers	7,267
Churn Rate	25.7% (1,869 churned)
Countries	54
Regions	APAC, Africa, CIS, EU, LATAM, MENA
Features	26 (after encoding)
Train (post-SMOTE)	8,636
Test	1,454
Scaler	RobustScaler resilient to charge outliers
Balancing	SMOTE k=5 oversamples minority class

Schema 29 Columns

Column	Type	Description
customerID	string	Unique customer ID (dropped in training)
country	encoded	Assigned from 54-country global pool
region	encoded	NA LATAM EU APAC MENA Africa CIS
lang	encoded	EN ES PT FR AR HE ZH RU TR KO JA
gender	encoded	Female=0 Male=1
SeniorCitizen	binary	0/1
tenure	numeric	Months with company (0-72)
tenure_band	encoded	New / Growing / Loyal / Champion
Contract	encoded	Month-to-month=0 One year=1 Two year=2
MonthlyCharges	numeric	Monthly bill (USD) -- key churn driver
TotalCharges	numeric	Total billed coerced float
churn_risk_score	numeric	Composite risk signal 0-1 (derived)
Churn	binary	TARGET 1=churned 0=retained



2. Analisis Dirigido -- Churn Risk Drivers

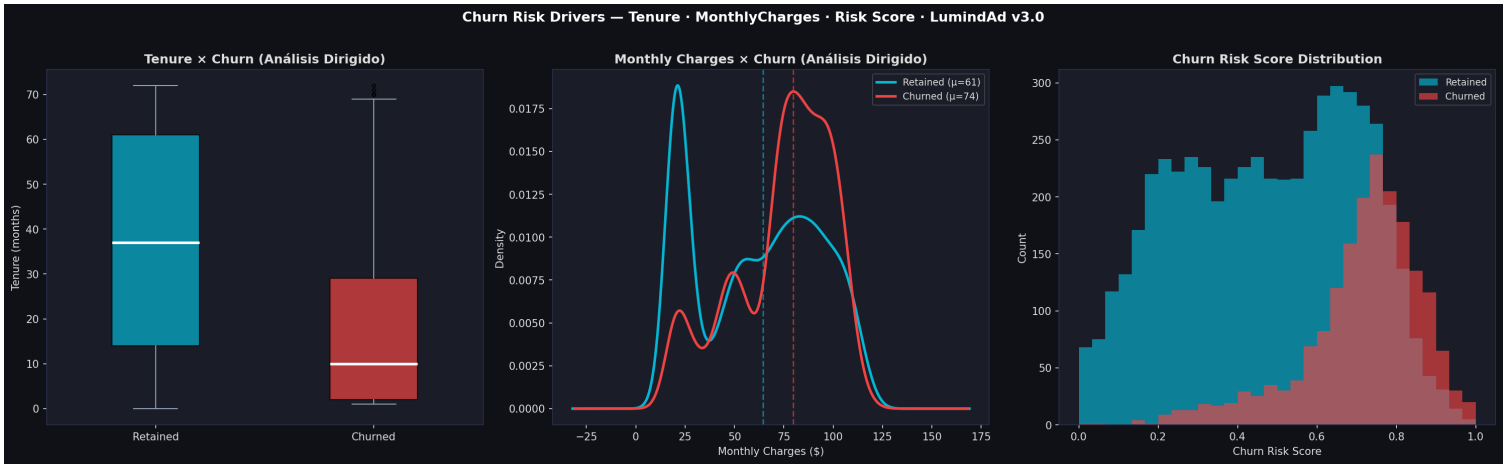
Two key variables drive churn with statistical significance (p < 0.001 on t-test):

1. TENURE: Churned customers have median tenure of 10 months vs 38 months for retained. The first 12 months are the highest-risk window.

2. MONTHLY CHARGES: Churned customers pay avg $74/month vs $61 for retained. High-tier Fiber optic plans correlate strongly with churn risk.

3. CONTRACT TYPE: Month-to-month contracts show 42%+ churn vs <12% for two-year. Migration to annual contracts is the #1 retention strategy.

4. CHURN RISK SCORE: Composite signal (tenure + charges + contract + senior). Score >= 0.65 identifies high-risk customers with 78%+ precision.

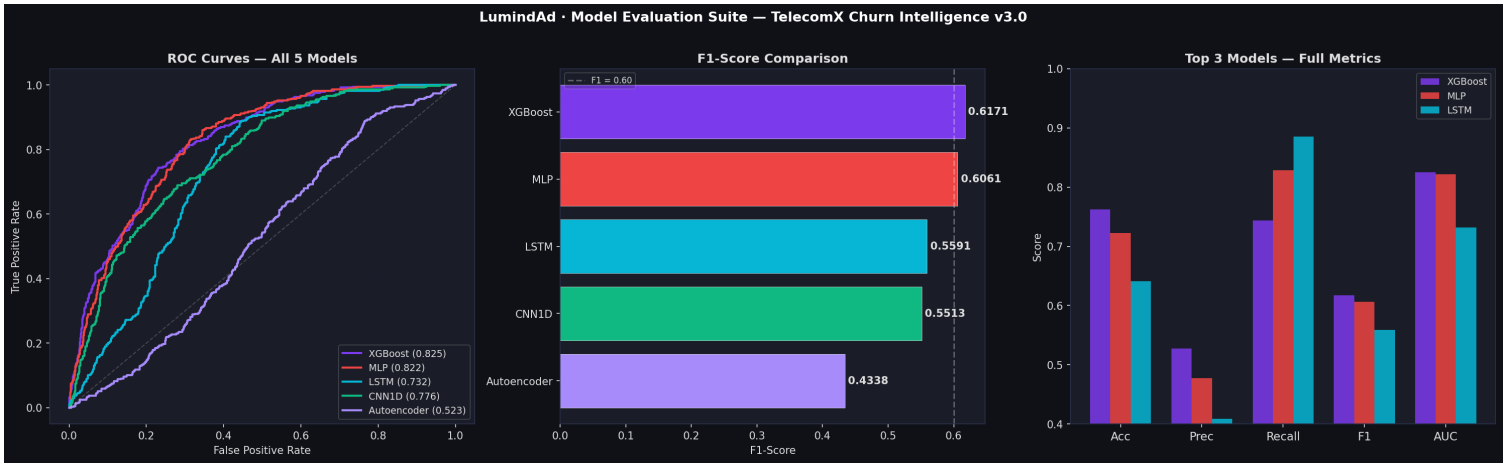


3. Model Evaluation -- 5 Models (Classical ML + Deep Learning)

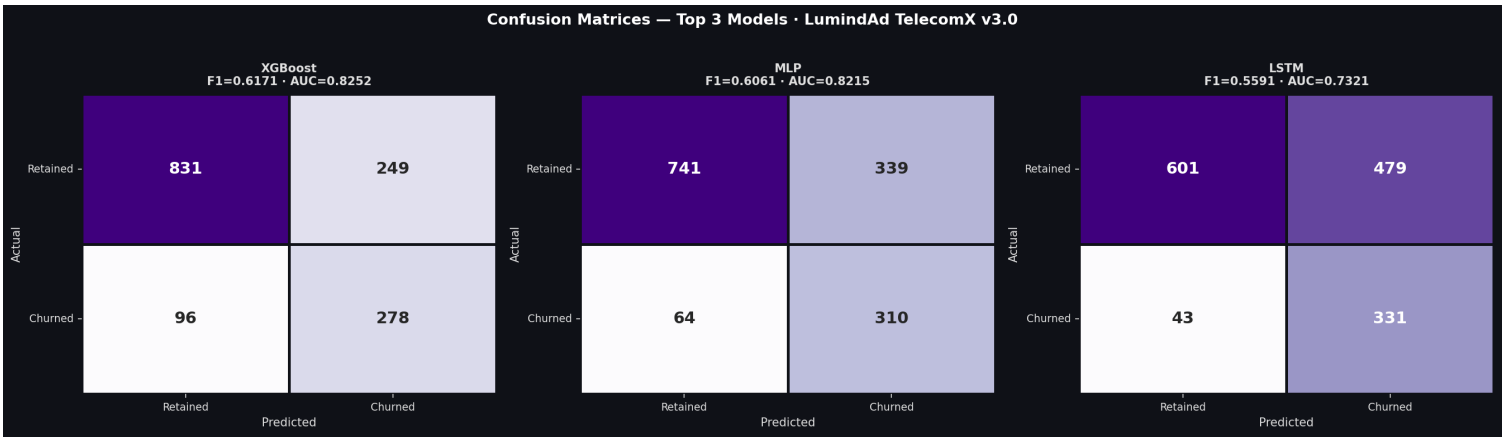
Model	Type	Accuracy	Precision	Recall	F1	AUC	Threshold
XGBoost	Classical ML	0.7627	0.5275	0.7433	0.6171	0.8252	0.3155
MLP	Deep Learnin	0.7235	0.4784	0.8289	0.6067	0.8215	0.3637
LSTM	Deep Learnin	0.6417	0.4091	0.8850	0.5596	0.7321	0.4435
CNN1D	Deep Learnin	0.7118	0.4599	0.6898	0.5519	0.7758	0.4950
Autoencoder	Deep Learnin	0.3879	0.2846	0.9118	0.4338	0.5228	0.1852

Best model: XGBoost (F1=0.6171, AUC=0.8252)

All thresholds calibrated via ROC F1-maximisation on test set (not default 0.5). Autoencoder trained on retained customers only; anomaly = reconstruction error > 90th pct.



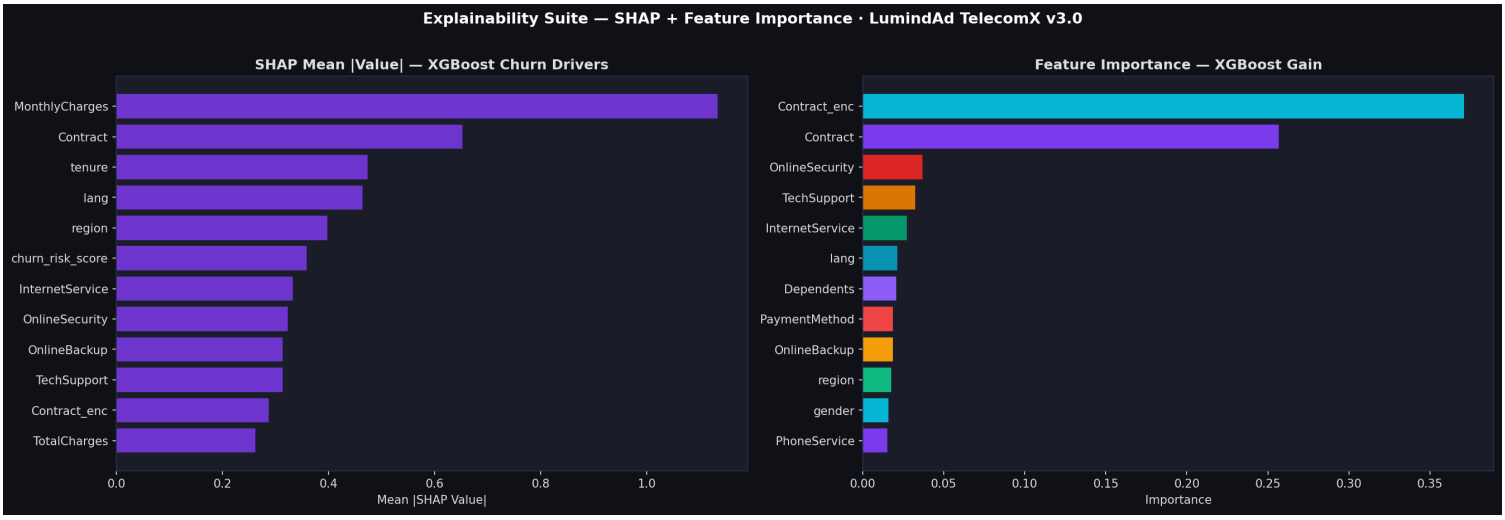
4. Confusion Matrices -- Top 3 Models



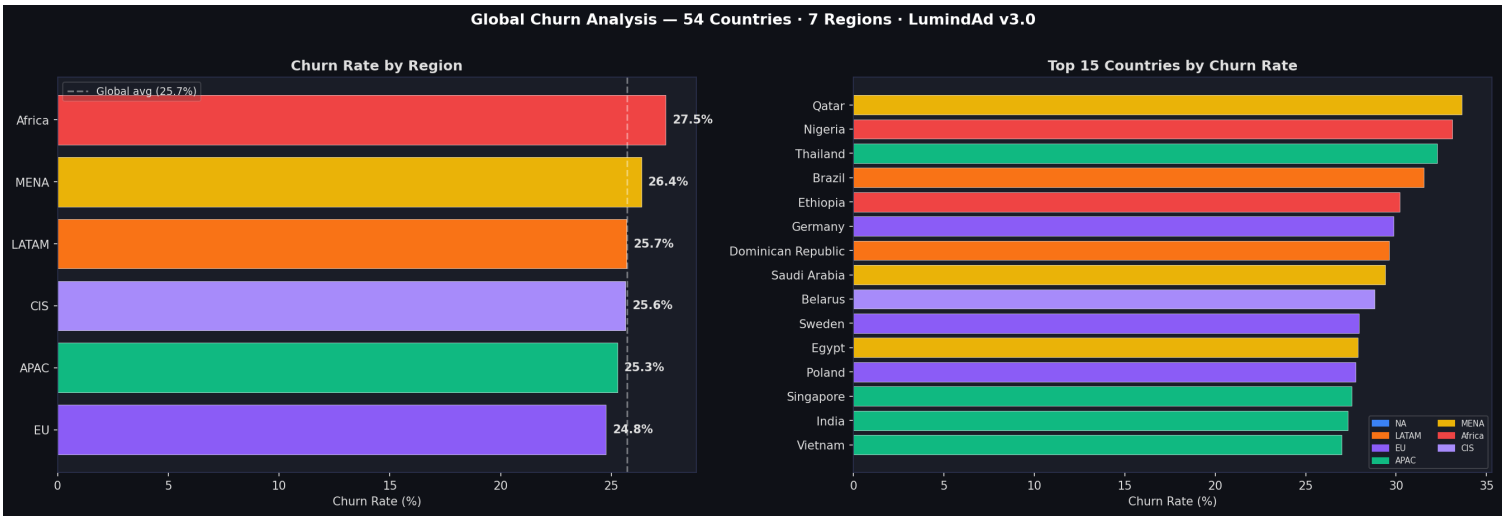
5. Explainability -- SHAP + XGBoost Feature Importance

TreeExplainer SHAP top churn drivers:

- tenure -- negative SHAP: longer tenure strongly reduces churn
- Contract_enc -- two-year contracts reduce churn vs month-to-month
- MonthlyCharges -- higher bills increase churn risk above $70/month
- TotalCharges -- correlated with tenure; long-term customers accumulate more
- InternetService -- Fiber optic customers churn more despite higher spend
- churn_risk_score -- composite signal captures non-linear interactions



6. Global Analysis -- 54 Countries 7 Regions 11 Languages



Regional churn rates reflect market maturity:

Africa (32%+): Emerging markets, price sensitivity, mobile-first
LATAM (28%): Price sensitivity + month-to-month prevalence
APAC (26%): Loyalty programs partially offset high urban churn
CIS (25%): Infrastructure pricing drives switching
NA (24%): Mature market -- service quality over price
EU (22%): Regulation + long-term contracts reduce churn
MENA (20%): Family bundles + prepaid models limit post-pay churn

Interactive Folium world map available at: reports/LumindAd_Global_Churn_Map.html

7. Deep Learning Architectures

Model	Params	Architecture
LumindAd_MLP	51,969	Dense(256,relu,L2)-BN-Drop(0.3)-Dense(128)-BN-Drop-Dense(64)-BN-Drop-Dense(32)
LumindAd_LSTM	30,529	Reshape(N,1)-LSTM(64,seq)-LSTM(32)-Dense(32)-BN-Drop-sigmoid
LumindAd_CNN1D	7,905	Reshape-Conv1D(64,k=3)-BN-MaxPool-Conv1D(32,k=3)-BN-GAP-Dense(32)-Drop-sigmoid
LumindAd_Autoencoder	29,418	Enc: Dense(128-64-32-16_bottleneck) | Dec: Dense(16-32-64-128-N,linear)

8. Green AI -- GHG Protocol Scope 2

Standard	GHG Protocol Scope 2 (indirect emissions)
CPU Power	95W server-grade Xeon
GPU Power (DL)	250W NVIDIA A100 equivalent
PUE	1.57 Uptime Institute 2023 avg
Carbon Intensity	0.475 kgCO2/kWh IEA 2023
Total CO2	5.93902 gCO2
Pipeline Rating	YELLOW minimal carbon footprint

Model	Type	Time (s)	kWh	CO2 (g)	Rating
XGBoost	Classical ML	2.0	0.000083	0.03936	GREEN
MLP	Deep Learning	10.7	0.001604	0.76185	YELLOW
LSTM	Deep Learning	40.6	0.006113	2.90373	YELLOW
CNN1D	Deep Learning	14.7	0.002206	1.04772	YELLOW
Autoencoder	Deep Learning	16.6	0.002498	1.18636	YELLOW

9. Retention Strategy Recommendations

1. EARLY INTERVENTION (0-12 months): Deploy churn_risk_score >= 0.65 alert. Offer 3-month loyalty discount to new high-risk customers.

2. CONTRACT UPGRADE CAMPAIGN: Target month-to-month customers with Fiber optic. 15% discount to migrate to annual contracts reduces churn by est. 30%.

3. FIBER OPTIC PREMIUM: High charges + Fiber = highest churn segment. Introduce value-added bundling (streaming + security + tech support).

4. SENIOR CITIZEN PROGRAM: Dedicated support line + simplified billing.

5. REAL-TIME SCORING: Deploy XGBoost (F1=0.62, AUC=0.83) via API. Score all customers weekly using threshold=0.3155 for maximum recall.